Curating clinical phenotypes in PHACE Syndrome to reveal disease mechanisms

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Kid's First Data Resource Portal

The Gabriella Miller Kid's First Data Resource Portal contains curated data about structural birth defects and childhood cancer, with the aim of expanding the understanding of these diseases, and making this data available to the clinical and research community.

PHACE Syndrome

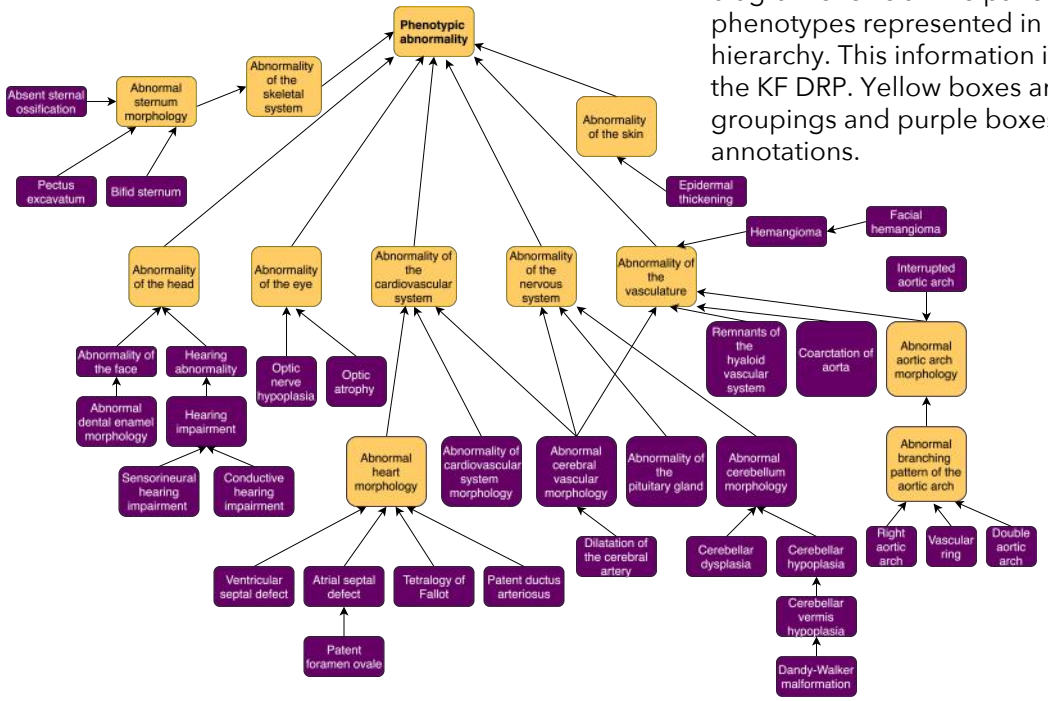
PHACE syndrome is a rare disorder of unknown etiology that is characterized by several structural birth defects (see box). The disease is difficult to diagnose due to the multiple overlapping phenotypes. Genetic variants underlying structural birth defects and associated complications in PHACE remain poorly understood, leading to difficulties treating and counseling patients and their families.

- ***P***osterior fossa brain malformations
- ***H***aemangiomas, face
- ***A***rtery abnormalities
- ***C***ardiac anomalies
- ***E***ye abnormalities

We curated patient phenotypes using the **Human Phenotype Ontology** (see panel below). This deep phenotyping can be used with analytics of genomic data to aid in diagnosis, variant prioritization and to ultimately improve diagnosis, precision medicine, and health outcomes for these patients. Here we report the identification of 15,752 rare, de novo variants based on gene sequencing of 100 PHACE patients, creating a need for additional computational tools for variant selection and prioritization.

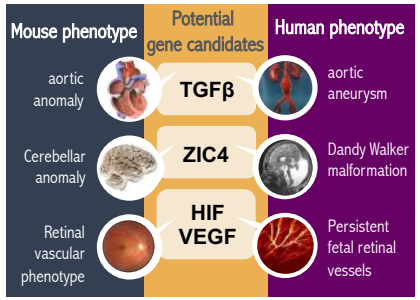
PHACE Phenotypes in HPO

PHACE phenotypes were curated for each patient using the HPO. This diagram shows all the patients' phenotypes represented in the HPO hierarchy. This information is available in the KF DRP. Yellow boxes are high level groupings and purple boxes represent annotations.



Next Steps

We will use the Monarch Initiative's tool, Exomiser, which uses an algorithm to find potential disease-causing variants from whole-exome or -genome sequencing data. Exomiser allows for comparison of gene to phenotype profiles from our curated patient phenotype data to model organism data. Exomiser will produce a ranked list of candidate variants for consideration of the underlying mechanisms of PHACE. The patient phenotypes can be compared against our curated Mendelian disease to phenotype associations (all encoded in HPO). Our intended outcome is to identify genes and variants that could help answer mechanistic questions, such as potential pathways or protein-protein interactions that may be affected in PHACE.



Hypothetical cross-species profile comparison. Phenotypes we see in PHACE syndrome are listed on the right, and corresponding phenotypes that are seen in existing mouse models and the associate genes are noted on the left and center. We expect that our cross-species analysis will identify candidate genes for PHACE syndrome in humans.

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RESOURCES:

Kid's First Data Resource Portal | portal.kidsfirstdrc.org/
Monarch Initiative | monarchinitiative.org
Human Phenotype Ontology | hpo.jax.org/app/